A-A-2348B
July 30, 1991
Superseding
A-A-2348A
October 19, 1987

COMMERCIAL ITEM DESCRIPTION

INDICATOR, DIAL, ACCESSORIES AND TEST SETS

The General Services Administration has authorized the use of this Commercial Item Description in preference to A-A-2348A. This description covers dial and test indicators, accessories and test stands.

1. CLASSIFICATION:

Type A - Spindle parallel to dial. Type A Indicators shall be supplied with a regular back (central vertical bracket).
Type B - Spindle perpendicular to dial from back of case. The holding rod shall be 1/4 inch diameter by 1-1/2 inches long, minimum, as measured from the bezel edge to the end of the rod. When specified, the item shall accommodate hole measuring accessories.
Type C - Universal adjustable contact point (test indicator). The contact point shall hold any position within a 220 degree arc. Type C indicators shall be quick acting, manual reversing type. The readings shall be unaffected by magnetic reading (non-magnetic type), as specified.

Style I - 0.0001 inch graduations
Style II - 0.0005 inch graduations
Style III - 0.001 inch graduations
Style V - 0.00005 inch graduations

Group 0 - Dial indicators having nominal bezel diameters from 1 inch up to and including 1-3/8 inch.
Group 1 - Dial indicators having nominal bezel diameters from above 1-3/8 inch up to and including 2 inches.
Group 2 - Dial indicators having nominal bezel diameters from above 2 inches up to and including 2-3/8 inches.
Group 3 - Dial indicators having nominal bezel diameters from above 2-3/8 inch up to and including 3 inches.
Group 4 - Dial indicators having nominal bezel diameters from above 3 inches up to and including 3-3/4 inches.

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

FSC 5210

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2. APPLICABLE DOCUMENTS:

The documents referenced in this Commercial Item Description (CID) shall be the issue in effect on the date of the Invitation for bids or request for proposal unless otherwise specified. These documents form a part of this CID to the extent specified. In the event of a conflict between this CID and a document referenced herein, this CID shall take precedence.

Federal Standards:
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

Copies of Federal Standards are available from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; and Seattle, WA.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Copies of military standards may be obtained from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

ANSI Standards:
ASME/ANSI B89.1.10M-1987, Dial Indicators (for linear measurements).

Application for copies should be addressed to the American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036.

3. SALIENT CHARACTERISTICS:

3.1 DESIGN

Dial indicators supplied in accordance with this description shall be constructed as to preclude entry of dust and dirt into the mechanism. Cases shall be of such strength and rigidity as to ensure free movement of the mechanism under normal shop conditions. Crystal shall be clear, transparent and manufactured with non-shattering materials. Contact points shall be of hardened steel or other wear resistant material with smooth uniform gaging surfaces. Bezels shall have knurls or serrations around the periphery. Dial indicators shall comply with the applicable requirements of Tables I, II, III and IV. Dial Indicators and components shall conform to ANSI B89.1.10M requirements for interchangeability.

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TABLE I

NSN (Type, Style)	Size group	Dial Reading	Spindle travel (inch)	Max. force to deflect spindle (grams)	
				25%	100%
5210-00-432-9235 (Type A, Style I)	2	0-5-0	0.025	140	170
5210-00-293-3016 (Type A, Style II)	2	0-25-0	0.125	105	120
5210-00-177-4191 (Type A, Style III)	4	0-50-0	0.500	95	150
5210-00-243-9649 (Type A, Style III)	2	0-100	1.000	115	260

TABLE II

NSN (Type, Style)	Bezel dia. inch	Dial Reading	Spindle travel (inch)	Max. force to deflect spindle (grams)	
				25%	100%
5210-00-277-8840 (Type B, Style III)	1-3/8 to 2-3/8	0-100	0.200	125	190
5210-00-377-6525 (Type B, Style III)	1-3/8 to 2-3/8	0-50-0	0.200	125	190

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TABLE III

NSN (Type, Style)	Bezel dia. inch	Dial reading	Lever travel min. (inch)	Max. Force to deflect lever upward (grams)	
				25%	100%
5210-00-273-9791 (Type C, Style I)	7/8 to 1-3/8	0-4-0	0.008	18	23
5210-00-223-9648 (Type C, Style III)	7/8 to 1-3/8	0-15-0	0.030	22	30

3.2 CONSTRUCTION

3.2.1 **Position.** The zero position of the dials shall be adjustable over a range of 360 degrees and the desired position fixed by a locking device or held by friction means between the case and bezel.

3.2.2 **Dial Hands.** The width of the tip shall be approximately the same as that of a graduation line on the dial face.

3.2.3 **Dial Indicator Range.** For Type A indicators, the minimum range shall be 2-1/2 revolutions of the indicating hand unless otherwise specified for applications requiring shorter or greater travel. Types B and C shall have a minimum range of 1 revolution of the dial hand. Dial indicators with longer than specified range are referenced in paragraph 8.4 of ANSI B89.1.10M.

3.2.4 **Physical Dimensions.** Refer to ANSI B89.1.10M, paragraph 8.2 section (d) for dimensions.

3.2.5 **Dial Faces.** The dial faces shall have sharp, distinct graduations and figures. Metric dial faces shall be yellow.

3.2.6 **Dial Markings.** Dial markings shall indicate the value of the least graduation, either inch or millimeter, and shall be in decimals, i.e. 0.001 in., not 1/1000 in.; or 0.01 mm, not 1/100 mm.

3.2.7 **Dial Numbering.** The dial numbering shall always indicate thousandths of an inch or hundredths of a millimeter, regardless of the value of dial marking.

3.3 ACCESSORIES. (Accessories shall be furnished with the dial indicators as specified in Table IV.)

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3.3.1 **Contact points.** Unless otherwise specified, standard 1/4 inch long contact points shall be supplied with Type A and B indicators. When specified, button, cone or ball contact points shall be supplied. Extra length contact points for Type C indicator shall be supplied when specified, providing that such contacts do not have an effect on the accuracy of the indicator.

3.3.2 **Hole attachment.** When specified, a hole measuring attachment with a depth capacity of 1-11/16 inch shall be supplied with Type A or B indicators. The device shall be capable of being securely attached to the spindle housing.

3.3.3 **Stanchion bases.** When specified, a stanchion base shall be supplied with the dial indicator and shall consist of the following components:

(1) A base and upright column with a minimum diameter of 0.465 inches by 8 inches long minimum with provision for attaching the column securely to the base.

(2) A dial holding rod 3/8 inch diameter by 8-1/4 inches long minimum with a thumbscrew clamp for attaching the indicator and a sliding swivel clamp for holding the indicator rod the upright column in any fixed position by tightening a thumbscrew or screws. The clamp and dial holding rod shall be capable of turning and swiveling 360 degrees on the upright column. The assembly shall contain a reducing bushing as specified.

(3) The standard stanchion base shall be 2-3/16 inches by 2 inches by 8-1/4 inches minimum with a longitudinal slot top and bottom for attaching the upright column.

3.3.4 **Arbor clamp.** When specified, an arbor clamp shall be furnished with the indicator. The clamp shall be essentially a "C" clamp with a 5/16 inch diameter by 5 inches long minimum forcing screw. The anvil with a "V" notch shall be opposite the forcing screw. The capacity of the clamp shall be not less than 1-1/4 inches. Shall be furnished with a tapped hole for mounting 5/16 inch diameter rod (one end threaded) to allow extension bar to be securely installed at a 90 degree angle to the clamp.

3.3.5 **Tool post.** Tool post mounting for Type C indicators shall be supplied when specified. The tool post mounting shall be either a holding bar with swivel clamp or universal friction holder with a universal swiveling arm and body clamp as specified.

3.3.6 **Test sets (tool post type), Type B indicators.** Type B indicator test sets shall be supplied with the following components:

(1) Type B indicator as specified.
(1) 1/4 inch by 5/16 inch clamp (complete sleeve)
(1) Hole attachment
(1) Arbor clamp
(1) Tool post holder, 3/8 inch by 3/4 inch by 5-3/4 inches long minimum, with 5/16 inch diameter by 3-1/4 inches, upright post. . .

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3.3.7 **Cases.** When specified, cases shall be supplied with the indicators or test sets. The case shall possess sufficient strength and durability to withstand the conditions encountered during normal shop use. It shall be designed to house compactly, and hold securely, all component parts, as specified, including the indicator.

TABLE IV

National Stock Number (NSN)	Style, Type, Accessories required
5210-00-432-9235	Type A, Style I, plunger type contact point action, angular base stop, standard stanchion base, case, jeweled bearings.
5210-00-293-3016	Type A, Style II, 1/4 inch long regular shape contact point, standard stanchion base without base stops, with offset arm, reducing bushing, tool post holder, case, jeweled bearings.
5210-00-177-4191	Type A, Style III, with revolution counter, jeweled bearings, telltale hand, plunger type contact point action, regular.
5210-00-243-9649	Type A, Style III, with revolution counter.
5210-00-277-8840 (Test Set)	Type B, Style III, plunger type contact point action, standard, button and ball contact points, hole attachment, arbor clamp, toolpost holder with upright post, clamp, case, jeweled bearings.
5210-00-377-6525	Type B, Style III, same accessories as 5210-00-277-8840 but with cone contact point in lieu of ball contact point, plunger type contact point action, case, jeweled bearings.
5210-00-273-9791	Type C, Style I, nonmagnetic type with ball contact point, lever gear assembly and hairspring made of nonmagnetic material and with holding bar with swivel clamp, cushioned movement, case, jeweled bearings.

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TABLE IV (continued)

National Stock Number (NSN)	Style, Type, Accessories required
5210-00-223-9648	Type C, Style III, ball contact point, universal friction tool post holder with universal swivel arm with clamping attachment.

3.4 MARKING

Each indicator shall be marked in a plain and permanent manner with the manufacturer's name or trademark, model number for source identification and state or country of manufacture. Accessories furnished in sets are not required to be marked with this information, however a label shall be attached on the inside lid of the set listing all accessories and part numbers (if applicable).

3.5 WORKMANSHIP

Dial indicators, accessories and test sets shall be free from any defect that impairs durability, serviceability and appearance. All moving parts shall operate freely and without restriction, and contact points shall be tightened sufficiently enough to prevent loosening during normal operation.

3.6 INSTRUCTIONS

Unless otherwise specified, the manufacturer shall furnish one copy of operating, calibration and maintenance instructions with each dial indicator.

3.7 METRIC PRODUCTS

Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED STD-376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.8 REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered material in accordance with public law 94-580, as amended to the maximum extent possible.

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4. QUALITY ASSURANCE PROVISIONS:

Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.1 Classification of Inspection

4.1.1 First Article Inspection. When specified in the contract or purchase order, the contractor shall perform all examinations and tests listed under paragraphs 4.3 and 4.4 except that the cycle test shall be conducted for 500,000 cycles. In addition, the maximum force required to deflect the spindle shall not exceed the requirements contained in Tables I, II and III, as applicable. This measurement shall be determined using a force measuring instrument.

4.1.1.1 Dial indicators shall be tested for accuracy, repeatability, and the force required to deflect the spindle or lever using the following materials and procedures.

Cycle testing shall be performed using the following materials:

Gage blocks - Range 0.10 to 4.0 inches
Accuracy: Grade 2 (GGG-G-15)

Microcalibrator (Federal Products Model 400B-5, 400B-25, or equal).

Mounting hardware - Suitable to mount all types of indicators to Microcalibrator.

Force Measuring Equipment.

Exerciser shall be a motor operated device used to cycle one or more dial indicators or other similar devices through their full range for the purpose of determining reliability. Design should be such as to allow dial indicators to be exercised with either a radial or axial load as specified in paragraph 4.4.1 at a rate of no greater than 50, but no less than 25 cycles per minute. A counter is required to verify the number of cycles. When the axial load test method is used, the flat spring shall be spring steel or equal mounted to bear on the cam between cam and contact and of sufficient thickness to prevent any lateral loads. Physical size may vary between

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exercisers, but the spring must maintain contact with the cam throughout the full rotation of the cam when axial testing is used.

Cycle testing shall be tested using the following procedures:

Set up exerciser with proper cams and positioner for the indicator being tested. Establish the type action, radial or axial, by interfacing a flat spring between cams and dial indicator point as required. Position and clamp the dial indicator so that the contact is against the cam or spring and the full range of the indicator is exercised in each revolution of the cam. Preset the counter to the desired cycles and run the cycle test. The cycle test must be performed for the specified cycles positioned in either the vertical or horizontal position. Cycle speed is set at a minimum of 25 full scale deflections per minute and a maximum of 50 full scale deflections per minute.

The cycle test must be run continuously without repair. Cleaning and/or oiling of the spindle after every 10,000 cycles is acceptable. No internal repair or cleaning can be done anytime during testing.

4.2 Sampling. A random sample of indicators and/or accessories shall be selected in accordance with MIL STD 105, from each lot of material offered for inspection.

4.2.1 Sampling for examination of visual and dimensional characteristics shall be at inspection level I, the acceptable quality level (AQL) shall be 2.5 percent expressed in terms of percent defective.

4.2.2 Sampling for quality conformance tests shall be at inspection level S-3, the acceptable quality level (AQL) shall be 4.0 percent expressed in terms of percent defective.

4.2.3. Quality Conformance Inspection. Samples from each production lot offered for inspection shall be selected, visually and dimensionally examined for all requirements of Section 3 of this CID and tested in accordance with paragraph 4.4.

4.3 Visual and Dimensional examination. Each sample unit, selected in accordance with paragraph 4.2.1 shall be examined for conformance with ANSI B89.1.10M-1987, paragraphs 4 and 5 and Tables I, II, III and IV of this CID, as applicable.

4.4 QUALITY CONFORMANCE TESTING

4.4.1 Cycle test. Using the test equipment and methodology described under paragraph 4.1.1.1, each sample unit, selected in

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accordance with paragraph 4.2.2, shall be cycled for 1,000 deflections. Type A and B indicators shall be tested in the axial modes and Type C indicators shall be tested in the radial mode. Units tested for quality conformance testing shall not be oiled and/or cleaned during the performance of the test.

4.4.2 Accuracy and repeatability testing. Each sample unit, selected in accordance with paragraph 4.2.2 shall be tested for accuracy and repeatability in accordance with the procedures contained in ANSI B89.1.10M-1987. These tests shall be performed prior to and after performance of the cycle test.

4.5 Examination for preparation for delivery. A random sample shall be inspected from each lot and examined for conformance with the preservation, packaging, packing, labeling and marking required in the contract or order. Samples shall be selected in accordance with Inspection level S-2, AQL 6.5.

5. PREPARATION FOR DELIVERY:

The preservation, packaging, packing, labeling and marking shall be as specified in the contract or order.

6. NOTES:

6.1 Intended use. Types A and B dial indicators with accessories are used principally on inspection fixtures and for aligning work pieces on machine tools such as lathes, milling machines, shapers, etc. Type C dial indicators are intended for centering and aligning work pieces on machine tools such as jig borers, lathes, boring mills, etc; and also checking dimensions between surfaces, center line of holes and keyways and runout shafts in inspection work.

MILITARY COORDINATING ACTIVITY:	PREPARING ACTIVITY:
Army - GL	GSA/FSS

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